

LCHAI v2.0 — User Manual

Lung Cancer Histologic Analysis with AI *Explainable mutation prediction from H&E whole-slide images*

DISCLAIMER: LCHAI v2.0 is a research tool (THESIS_INTERNAL evidence). Mutation predictions use ABMIL + Fuzzy Choquet MIL. Inconclusive genes (AUROC < 0.70) require molecular testing. NOT for clinical diagnosis.

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1. System Overview

LCHAI v2.0 is an AI-powered decision-support system for lung adenocarcinoma (LUAD) histopathology. It analyzes whole-slide images (WSI) to:

- **Classify histological growth patterns:** lepidic, acinar, papillary, micropapillary, solid, cribriform

- **Predict oncogenic mutations:** TP53, EGFR, KRAS, STK11, KEAP1, RBM10
- **Provide six-level explainability:** attention maps, pattern overlays, SHAP decomposition, Choquet Shapley values, per-slide ablation, and fuzzy linguistic labels
- **Generate knowledge graphs** linking predictions to biomedical ontologies (NCIt, MONDO) and treatment associations
- **Produce AI-generated explanations** grounded in the knowledge graph with numbered PubMed citations

ML Pipeline Summary

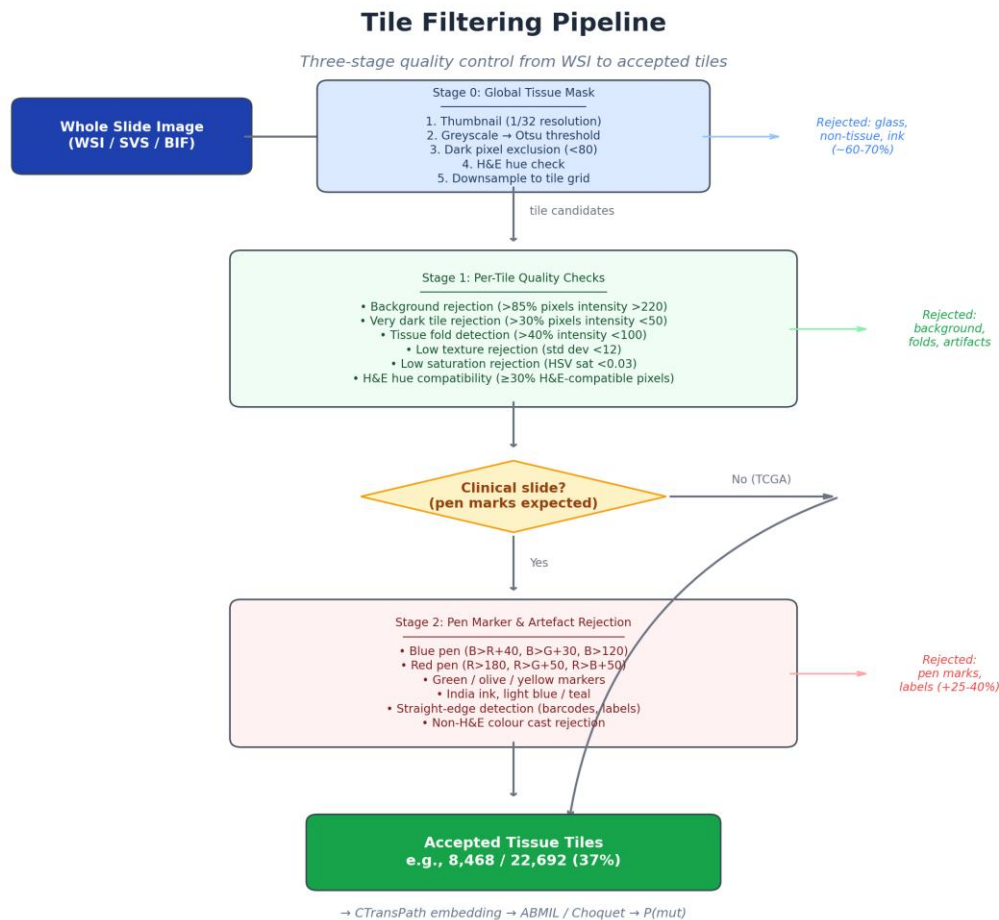


Figure 1: Three-stage tile filtering pipeline from WSI to accepted tissue tiles

1. **WSI Decode** — Opens the slide file (SVS, TIFF, or BIF) using OpenSlide
2. **Tile Filtering** — Three-stage quality control rejects glass, artifacts, pen marks (see Figure 1)
3. **CTransPath Inference** (GPU) — Each tile produces a 512-dimensional embedding

4. **Pattern Classification** — FuzzyArcLoss V2 classifies each tile into 1 of 6 histological patterns
 1. **Mutation Prediction** — Gene-specific models (ABMIL or Fuzzy Choquet MIL) predict mutation probability
 2. **Explainability** — Ablation, SHAP decomposition, Choquet Shapley values, and fuzzy linguistic labels
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2. Getting Started — Login

Navigate to the LCHAI system (<https://lchai.gptfy.biz> or <http://localhost:3000>). You will be redirected to the Keycloak authentication page.

LUNG CANCER HISTOLOGIC ANALYSIS WITH AI

Sign in to your account

Username or email

Password



Sign In

Figure 2: Keycloak login screen

Method	Description
Username / Password	Enter your assigned credentials (e.g., admin1 / admin1)
Google	Click “Google” to sign in with your Google account (OAuth2)

User Roles

Role	Access
Clinician	Analysis, Viewer, Graph tabs
Admin	All clinician access + Admin tab
Auditor	Read-only access to audit trail

3. Analysis Tab — Upload, Analyze, and Explain

The Analysis tab is the primary workspace. It integrates image upload, mutation prediction, and multi-level explainability into a single unified view.

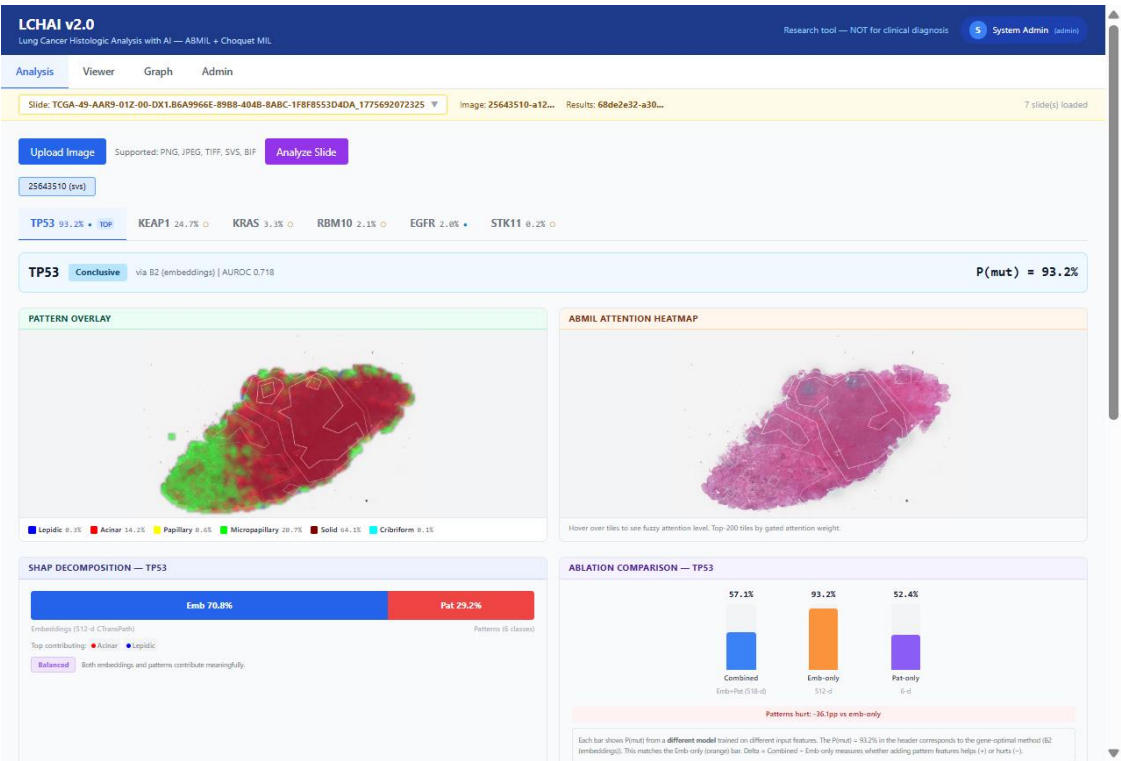


Figure 3: Analysis tab showing TP53 (93.2%, Conclusive). Gene sub-tabs at top, side-by-side overlays, SHAP with fuzzy label, and ablation comparison.

3.1 Uploading an Image

1. Click **“Upload Image”** (blue button)
2. Select a histopathology image file (PNG, JPEG, TIFF, SVS, BIF)
3. A progress bar shows upload percentage
4. The system creates a new case and selects the uploaded slide

Slide Selector: The yellow bar at the top shows all uploaded slides. Click the dropdown to switch.

3.2 Analyzing a Slide

1. Select a slide, then click **“Analyze Slide”** (purple button)
2. A processing modal shows real-time progress:

Stage	Description	Time
Decoding image	Opens WSI, creates thumbnail	5-10s
CTransPath inference	GPU processes tiles	10-30s
Mutation prediction	ABMIL/Choquet for 6 genes	5-10s
Ablation + permutation	Model comparison	10-20s
SHAP decomposition	Feature attribution	5-15s
Saving results	Store in database	2-5s

Click **“Cancel and discard”** to abort at any time.

3.3 Gene Sub-tabs

After processing, gene sub-tabs appear **sorted by descending P(mut)**. The highest-probability mutation is selected by default.

Each tab shows: - Gene name, P(mut) percentage, and prediction method - A filled circle (●) for Conclusive genes, empty circle (○) for Inconclusive - A “TOP” badge on the highest-probability gene

3.4 Pattern Overlay and ABMIL Attention Heatmap

Two images are displayed side by side:

Left: Pattern Overlay	Right: ABMIL Attention Heatmap
Tile-level color map of 6 growth patterns	Top-200 tiles by gated attention weight
Legend shows pattern names with composition %	Red = high attention
Hover: shows tile pattern name and %	Hover: shows fuzzy attention label (e.g., “High Attention, p97”)

Pattern Color Legend:

Color	Pattern
Blue	Lepidic
Red	Acinar
Yellow	Papillary
Green	Micropapillary
Dark red	Solid
Cyan	Cribriform

3.5 SHAP Decomposition with Fuzzy Balance Label

A stacked bar shows what percentage of the prediction comes from: - **Embeddings** (512-d CTransPath visual texture) — blue - **Patterns** (6-class histological classification) — red

A **fuzzy linguistic label** classifies the balance:

Label	Meaning
Embedding-Dominated	Patterns nearly irrelevant (<10%)
Embedding-Led	Patterns provide minor signal (10-25%)
Balanced	Both contribute meaningfully (25-55%)
Pattern-Led	Patterns are primary signal (55-70%)
Pattern-Dominated	Embeddings secondary (>70%)

3.6 Ablation Comparison

Three vertical bars compare P(mut) from **three independently trained models**: - **Combined** (blue): embeddings + patterns (518-d) - **Emb-only** (orange): embeddings alone (512-d) - **Pat-only** (purple): patterns alone (6-d)

A delta indicator shows whether patterns help (+) or hurt (-).

Important: These are three different models, not components of one prediction. The P(mut) in the header comes from the gene-optimal method, which may differ from any bar.

3.7 AI-Generated Explanation

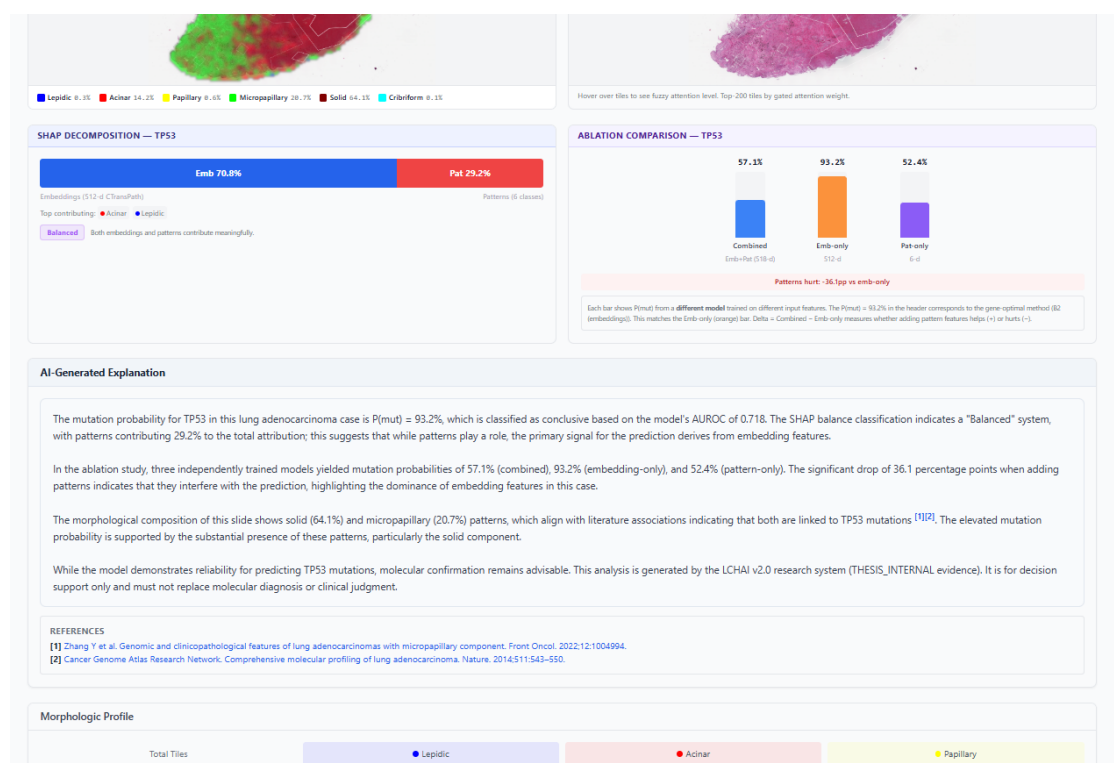


Figure 4: AI-generated explanation with SHAP fuzzy label, ablation interpretation, literature citations [1][2], and PubMed references

The system generates a structured clinical narrative (8–12 sentences) that: - States $P(\text{mut})$ and whether it is conclusive or inconclusive - Describes the SHAP balance using the system's fuzzy label - Interprets the ablation comparison - Compares the slide's patterns to literature expectations with **numbered PubMed citations** [1], [2] - Includes the thesis disclaimer

All fuzzy labels and clinical facts are **system-controlled** — the LLM cannot invent intensity terms or clinical associations.

References: Below the explanation, clickable PubMed links for each cited paper.

3.8 Fuzzy Choquet Analysis (Conditional)

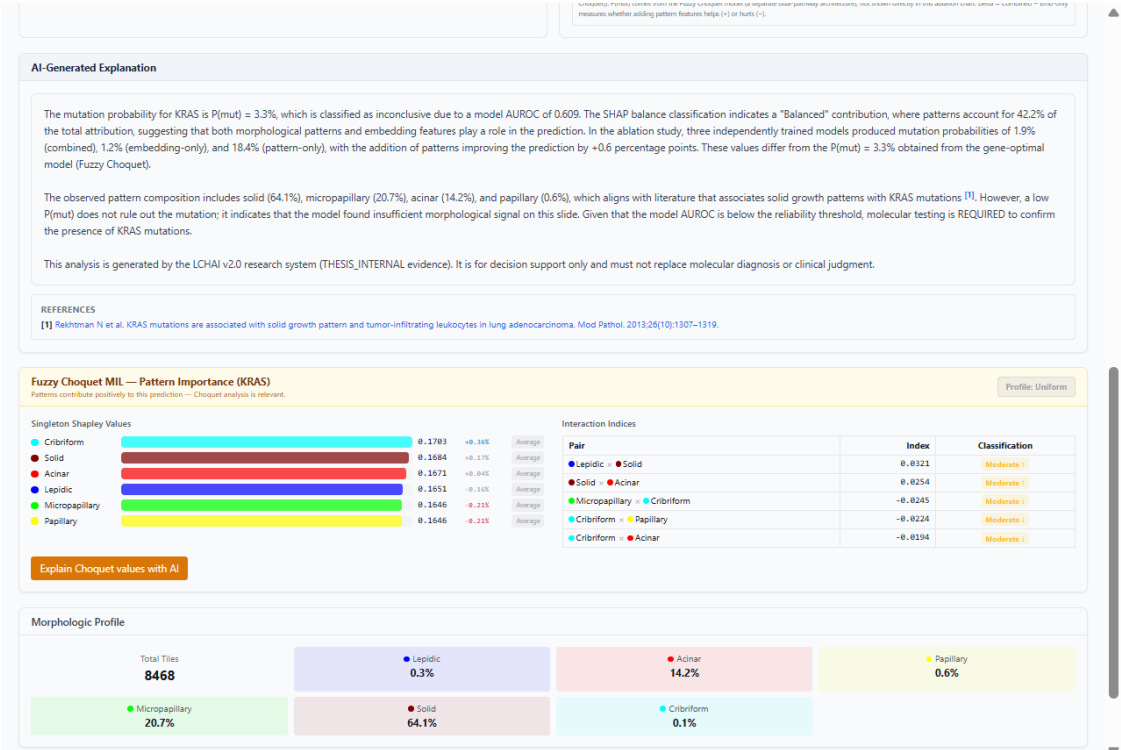


Figure 5: Choquet section for KRAS with “Profile: Uniform” badge, per-pattern Shapley labels, and interaction indices with fuzzy classifications

This section appears **only when patterns contribute positively** (Combined > Emb-only).

Component	Description
Profile badge	Fuzzy classification of Shapley spread (e.g., “Uniform”, “Near-Uniform”)
Shapley value bars	Per-pattern importance with individual fuzzy labels (e.g., “Average”)
Interaction indices table	Pairwise synergy/redundancy with fuzzy intensity (e.g., “Moderate Synergy ↑”)
Explain with AI button	LLM interpretation using the system-computed fuzzy labels

3.9 Morphologic Profile

Grid showing total tile count and percentage breakdown of all 6 histological patterns.

4. Viewer Tab — Image Exploration

The Viewer tab provides an interactive image viewer with zoom, pan, and overlay layers.



Figure 6: Viewer tab with pattern overlay layer

Layer Controls

Button	Layer	Description
1 Original	H&E image	The original histopathology image
2 Patterns	Pattern overlay	Color-coded pattern classification per tile
3 Attention	ABMIL attention	Heatmap with white contour lines
4 Combined	Both	Pattern colors + attention contours

Navigation

Action	How
Zoom in/out	Scroll wheel or +/- keys
Pan	Click and drag
Reset view	Click “Fit” or press 0
Switch layers	Press 1, 2, 3, 4

Hover Tooltips

On Pattern and Combined layers, hovering over a tile shows the **pattern name** with a color indicator.

Active Learning (Pattern Correction)

In Pattern/Combined layers, click **“Correction Mode”** to: 1. Draw a lasso around misclassified tiles 2. Select the correct pattern from the dropdown 3. Submit corrections for delta retraining

5. Graph Tab — Knowledge Graph

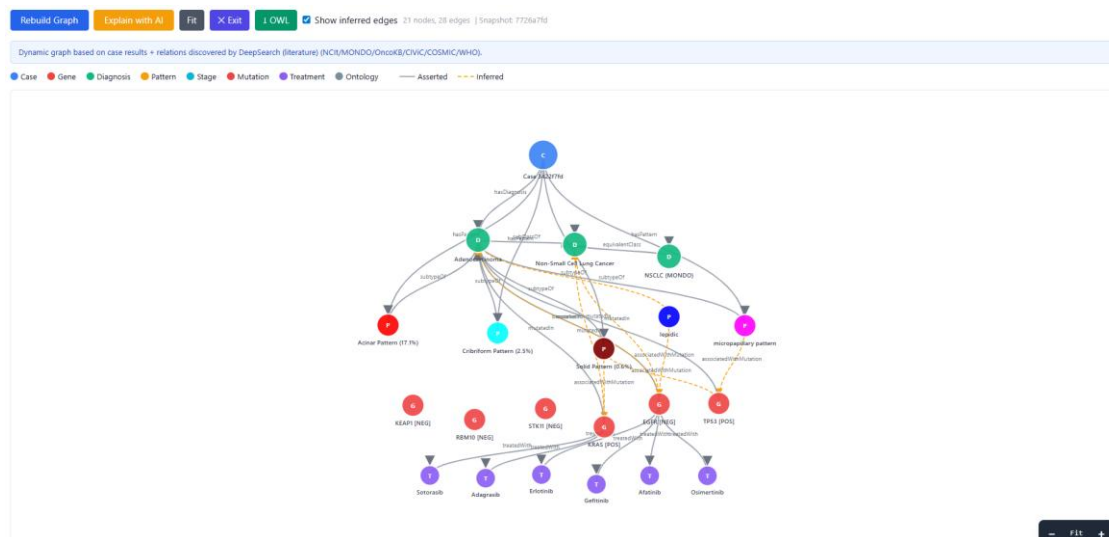


Figure 7: Knowledge graph linking patterns, genes, treatments, and ontology concepts

Building the Graph

Click **“Rebuild Graph”** to generate a case-specific knowledge graph from: - Predicted mutations and patterns - NCI and MONDO ontology relationships - Treatment associations (OncoKB/FDA guidelines) - DeepSearch-discovered literature relations

AI Explanation

Click **“Explain with AI”** to generate a natural language summary in your selected language.

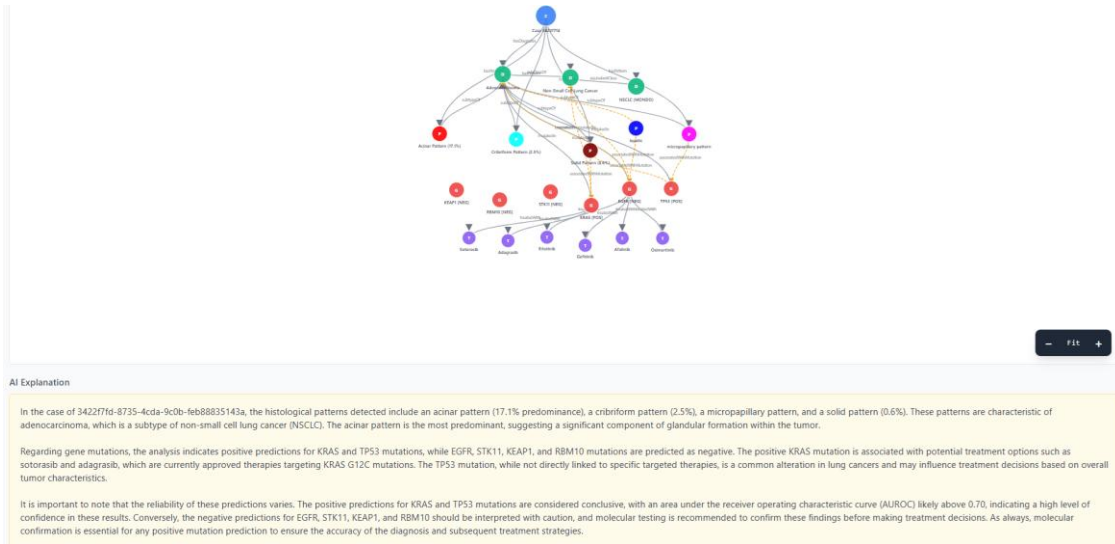


Figure 8: LLM-generated graph explanation

6. Admin Tab — System Configuration

Accessible only to users with the **admin** role. Contains seven sub-tabs:

Parameters

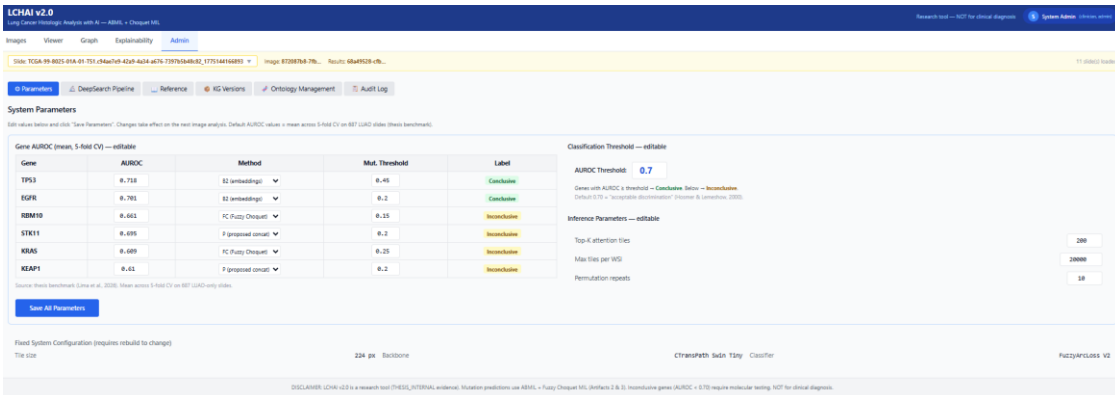


Figure 9: System parameters — AUROC values, methods, thresholds

Parameter	Description	Default
Gene AUROC values	Mean 5-fold CV AUROC per gene	0.609–0.718
Method per gene	Optimal model (B2, P, or FC)	From thesis Finding 2
AUROC Threshold	Conclusive/Inconclusive boundary	0.700
Max tiles per WSI	Tile extraction limit	20,000

Fuzzy Labels

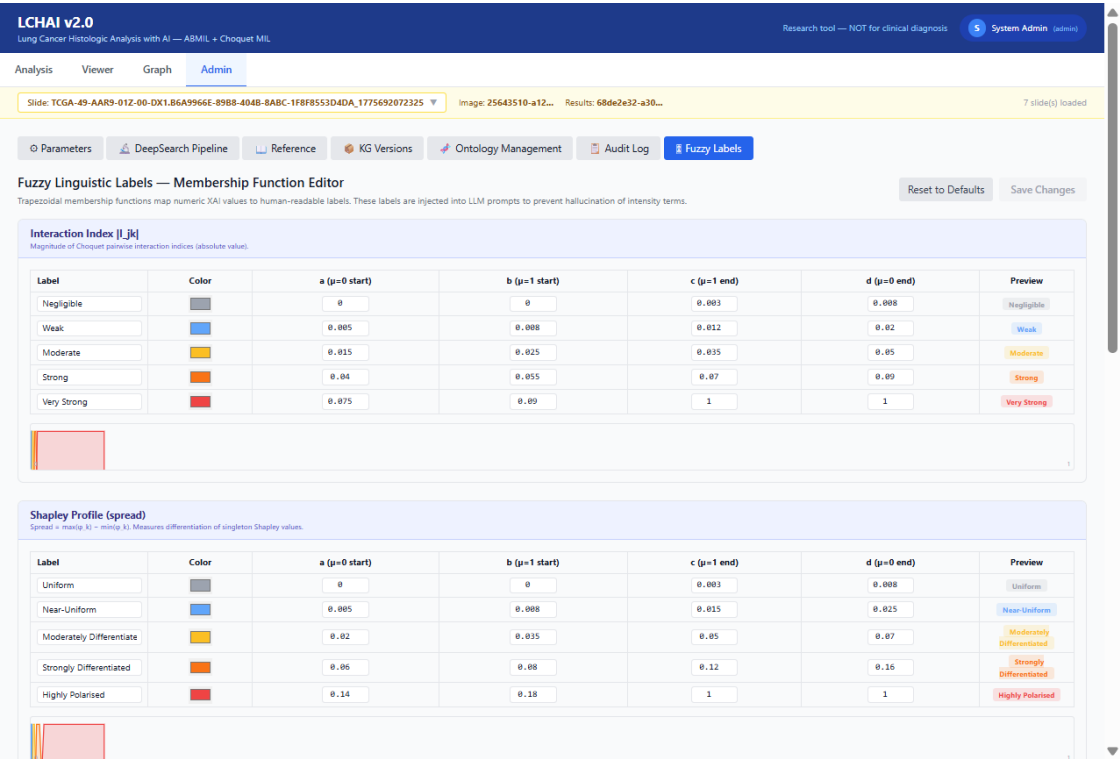


Figure 10: Fuzzy linguistic label editor with trapezoidal membership functions

Edit the five fuzzy scales that classify XAI outputs: 1. **Interaction Index** — Negligible to Very Strong 2. **Shapley Profile** — Uniform to Highly Polarised 3. **SHAP Balance** — Embedding-Dominated to Pattern-Dominated 4. **Shapley Individual** — Average to Strongly Above 5. **ABMIL Attention Level** — Very Low to Very High

Each scale shows editable trapezoidal parameters [a, b, c, d] with real-time SVG preview. Changes take effect immediately in the Analysis tab.

Other Sub-tabs

Sub-tab	Purpose
DeepSearch Pipeline	Automated literature search for new gene-pattern-treatment relations
Reference	Clinical reference tables and method formulas
KG Versions	Knowledge graph snapshot management
Ontology Management	NCIt/MONDO ontology update proposals
Audit Log	System event history with user, case, action, timestamp

7. User Profile and Settings

Click your **name** in the top-right corner to access:

- **Profile:** Name, email, role badges
 - **Explanation language:** English, Espanol, Deutsch, Francais, Portugues
 - Affects AI-generated explanations in the Analysis and Graph tabs
 - **Logout:** End session
-

8. Supported Image Formats

Format	Extension	Notes
SVS	.svs	Recommended. Fastest via OpenSlide
TIFF	.tif, .tiff	Flat TIFFs auto-converted to pyramidal
BIF	.bif	Ventana/Roche. Auto-converted via pyvips
PNG	.png	For small sections
JPEG	.jpg, .jpeg	For small sections

9. Understanding the Results

Mutation Probability

P(mut)	Interpretation
> 70%	Strong evidence — confirm with molecular testing
50-70%	Moderate evidence — molecular testing recommended
30-50%	Weak signal — molecular testing needed
< 30%	Likely wild-type

Conclusive vs Inconclusive

- **Conclusive** (AUROC ≥ 0.70): Reliable prediction. The model has demonstrated acceptable discrimination.
- **Inconclusive** (AUROC < 0.70): Limited accuracy. Molecular testing is strongly recommended.

Gene-Specific Associations

Gene	Pattern association	Treatment implications
TP53	Solid, micropapillary	No targeted therapy; immunotherapy in selected cases
EGFR	Lepidic, papillary	Osimertinib, erlotinib, gefitinib, afatinib

Gene	Pattern association	Treatment implications
KRAS	Solid (mucinous not in taxonomy)	Sotorasib, adagrasib (G12C-specific)
STK11	Variable (immune-cold)	May predict IO resistance
KEAP1	Diffuse	NRF2 pathway; affects IO response
RBM10	Variable	No targeted therapy; splicing biology

10. Fuzzy Linguistic Labels

LCHAI uses **trapezoidal fuzzy membership functions** to classify numeric XAI outputs into human-readable intensity terms. These labels are:

- Displayed as badges in the Analysis tab
- Injected into LLM prompts to **prevent hallucination** of intensity terms
- Editable by administrators via Admin > Fuzzy Labels

The Five Scales

Scale	Input	Labels
SHAP Balance	Pattern contribution %	Emb-Dominated → Emb-Led → Balanced → Pat-Led → Pat-Dominated
Shapley Profile	Spread (max-min)	Uniform → Near-Uniform → Mod. Diff. → Strongly Diff. → Polarised
Shapley Individual	% deviation from uniform	Average → Slightly → Moderately → Strongly Above/Below
Interaction Index	$ I_{jk} $	Negligible → Weak → Moderate → Strong → Very Strong
Attention Level	Tile percentile	Very Low → Low → Moderate → High → Very High

Anti-Hallucination

The fuzzy labels create a **dual anti-hallucination mechanism**: 1. **Knowledge Graph** constrains *what* the LLM says (clinical facts with PMID provenance) 2. **Fuzzy labels** constrain *how* it says it (intensity terms computed deterministically)

This ensures the AI explanation matches exactly what is displayed in the interface.

11. Troubleshooting

Problem	Solution
Blank screen after login	Refresh (F5) or clear browser cache
Upload stuck at 0%	Check internet connection
Processing too slow	Normal ~1min GPU, ~5min CPU. Check Admin > Max tiles
No results after processing	Select the correct slide in the dropdown
BIF file fails	Large BIFs auto-convert. Check worker logs
Admin tab not visible	Requires admin role
Explanation in wrong language	Change in user menu (top-right)
Attention hover not showing labels	Slide needs re-processing with current pipeline
Choquet section not appearing	Only shows when patterns help (Combined > Emb-only)

Performance Tips

- Use **SVS format** for large slides
- Keep **Max tiles at 20,000** for balanced accuracy/speed
- **GPU acceleration** reduces processing from ~5 min to ~1 min
- Close other tabs when uploading large files (>2 GB)

LCHAI v2.0 — Servio Fernando Lima Reina, PhD Computer Science, University of Fribourg, Switzerland For research use only. Always confirm AI predictions with molecular testing.